Financial Economics – Lesson 4: Welfare and Multidate Markets

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Outline

Welfare
- Pareto Dominance
- Social Planner Problem
- Welfare Theorems

Multidate model
- Multidate market
- Multidate valuation functional

Pareto dominance

Consider N agents with utilities $U^1(\cdot), \ldots, U^N(\cdot)$, the allocation $\{\mathbf{c}^n\}$ is *feasible* if $\sum_n \mathbf{c}^n \leq \mathbf{w}$: $\sum_n c_s^n \leq w_s\,,\ s = 0, 1, \ldots, S$.

Now consider a different allocation $\{\bar{\mathbf{c}}^n\}$, then if

- $\forall n\ U^n(\bar{\mathbf{c}}^n) \geq U^n(\mathbf{c}^n)$ the allocation $\{\bar{\mathbf{c}}^n\}$ *weakly pareto dominates* $\{\mathbf{c}^n\}$,
- $\forall n\ U^n(\bar{\mathbf{c}}^n) \geq U^n(\mathbf{c}^n)$ and $\exists j$ s.t. $U^j(\bar{\mathbf{c}}^j) > U^j(\mathbf{c}^j)$ the allocation $\{\bar{\mathbf{c}}^n\}$ *pareto dominates* $\{\mathbf{c}^n\}$.

Definition

An allocation $\{\mathbf{c}^n\}$ is *pareto optimal* if there exists no feasible allocation that dominates it.

The utility of each agent is compared only with itself: improve with respect to a *welfare function*.

Social Planner Problem

Consider the utilities $U^1(\cdot), \ldots, U^N(\cdot)$ and endowments $\mathbf{w}^1, \ldots, \mathbf{w}^N$, the social planner problem is

$$\max_{\{\mathbf{c}^1,\ldots,\mathbf{c}^N\}} \quad U = \sum_n \eta^n U^n(\mathbf{c}^n) \quad \text{s.t.}$$

$$\begin{cases} \sum_n \mathbf{c}^n \leq \mathbf{w}\,, \\ \\ \mathbf{c}^n \geq \mathbf{0} \quad \forall n\,, \end{cases}$$

where the η's are positive coefficients. This problem consists in the maximization of a *welfare function* U over the set of feasible allocations. The maximization is over a compact and convex set so that

- if the U's are continuous a maximum exists,
- if the U's are quasi-concave the maximum is unique.

Solution of Social Planner Problem

The solution of the social planner problem is clearly a Pareto optimum: the existence of a different allocation that provides higher utility to at least one agent would increase the social planner objective function.

Candidate solutions maximize

$$\mathcal{L} = \sum_n \eta^n U^n + \sum_s \lambda_s \left(w_s - \sum_n c_s^n \right) + \sum_{n,s} \bar{\lambda}_s^n c_s^n$$

and the system of F.O.C + slackness conditions read

$$\begin{cases} \partial_s U^n = \dfrac{\lambda_s}{\eta_n} - \dfrac{\bar{\lambda}_s^n}{\eta^n} & s \in \{0, 1, \ldots, S\}, n \in \{1, \ldots, N\} \\ (w_s - \sum_n c_s^n)\lambda_s \doteq 0 & s \in \{0, 1, \ldots, S\} \\ \bar{\lambda}_s^n c_s^n \doteq 0 & s \in \{0, 1, \ldots, S\}, n \in \{1, \ldots, N\} \end{cases}$$

Economic Equilibrium

In economic equilibrium for $n \in \{1, \ldots, N\}$ solutions solve

$$\begin{cases} \partial_s U^n = \mu_s^n - \bar{\mu}_s^n & s \in \{0, 1, \ldots, S\} \\ \mu_0^n p_j = \sum_s \mu_s^n x_{j,s} & j = \{1 \ldots J\} \\ (w_s^n + \mathbf{h}^n \mathbf{x}_{.,s} - c_s^n)\mu_s^n \doteq 0 & s \in \{0, 1, \ldots, S\} \\ \bar{\mu}_s^n c_s^n \doteq 0 & s \in \{0, 1, \ldots, S\} \end{cases}$$

with $\mathbf{x}_{.,0} = -\mathbf{p}$. If the market is complete and utility is increasing: $\mu_0^n > 0$, $\mu_s^n = q_s \mu_0^n > 0$ ($q_0 = 1$) and substituting

$$\partial_s U^n = q_s \mu_0^n - \bar{\mu}_s^n s \in \{0, 1, \ldots, S\} \ .$$

When $\mathbf{h} = 0$, correspond to the social planner problem with $\lambda_s = q_s$ and $\eta^n = 1/\mu_0^n$.
Thus social planner problem $\Longleftrightarrow$ economic equilibrium.

Pareto optimal I

With positive consumption the gradients of the utility functions $\nabla U^n = (\partial_0 U^n, \partial_1 U^n, \ldots, \partial_S U^n)$ are parallel for all agents:

$$\nabla U^n \;//\; (1, q_1, \ldots, q_S) \;//\; (\lambda_0, \mu_1, \ldots, \lambda_S)$$

for both the social planner problem and the economic equilibrium.

Let $\{\mathbf{c}^{*\,n}\} >> 0$ be an internal Pareto optimal allocation. Then, for any couple of agents n and m and for any variation $\delta\mathbf{c}$ in the allocation

$$\text{if } U^n(\mathbf{c}^{*\,n} + \delta\mathbf{c}) \geq U^n(\mathbf{c}^{*\,n}) \text{ then } U^m(\mathbf{c}^{*\,m} - \delta\mathbf{c}) \leq U^m(\mathbf{c}^{*\,m})\,.$$

Pareto Optimal II

For small variations $\delta\mathbf{c}$ we can take the differential version

$$\text{if} \quad \nabla U^n(\mathbf{c}^{*\,n})\delta\mathbf{c} \geq \mathbf{0} \quad \text{then} \quad -\nabla U^m(\mathbf{c}^{*\,m})\delta\mathbf{c} \leq \mathbf{0}\,,$$

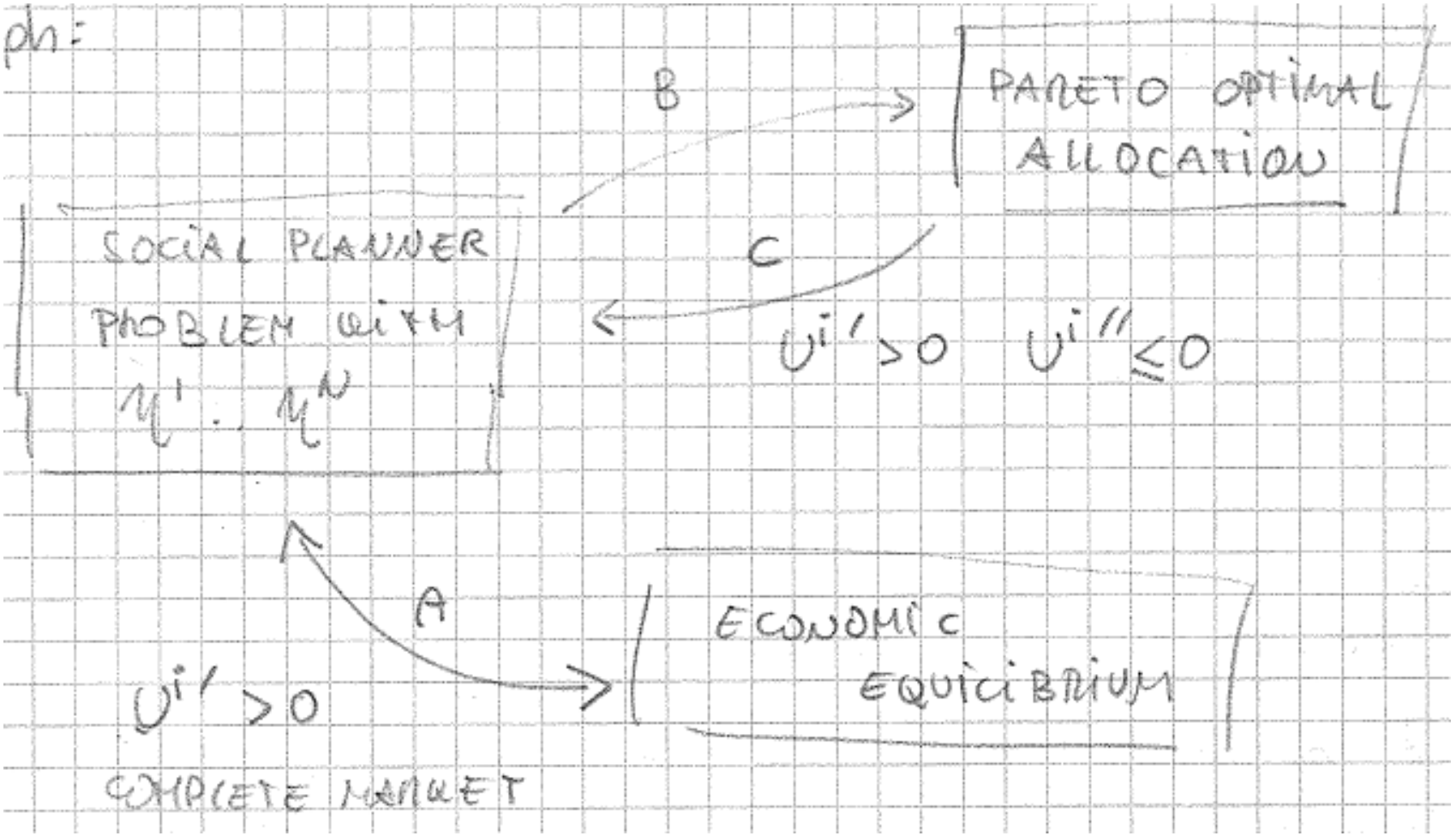
which means that

$$\text{if} \quad \nabla U^n(\mathbf{c}^{*\,n})\delta\mathbf{c} \geq \mathbf{0} \quad \text{then} \quad \nabla U^m(\mathbf{c}^{*\,m})\delta\mathbf{c} \geq \mathbf{0}\,.$$

This condition implies that ∇U^n and ∇U^m are collinear.

Thus it is $\partial_s U^n = a^n \Delta_S$, $a^n > 0$. If the utility functions are increasing, this is equivalent to a non-zero consumption solution of the social planner problem with $\eta^n = 1/a^n$. If in addition the utility functions are concave, these F.O.C. correspond to a maximum.

Welfare Theorems (graph)



Welfare Theorems

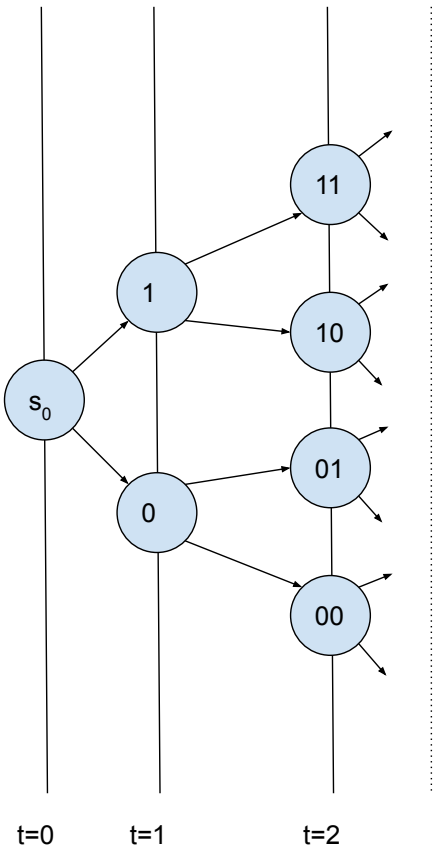
Theorem (First Welfare Theorem)

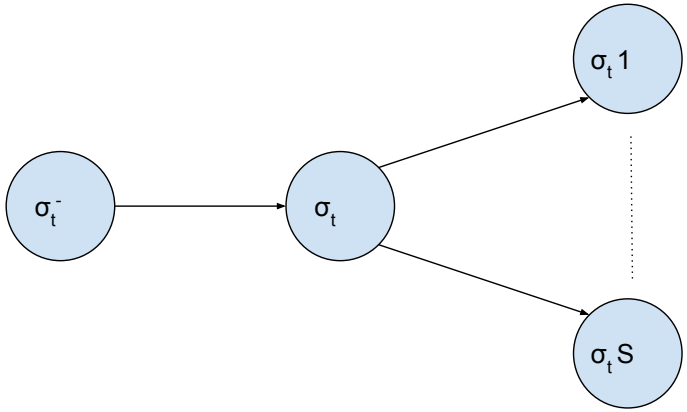
Economic Equilibrium $\underset{U^{n\prime}>0\,,\ \textit{complete market}}{\longrightarrow}$ *Social Planner Optimum for some* $\boldsymbol{\eta} > 0$ $\underset{U^{n\prime}>0\,,\ U^{n\prime\prime}\leq 0}{\longrightarrow}$ *Pareto Optimal Allocation.*

Theorem (Second Welfare Theorem)

Pareto Optimal Allocation $\underset{U^{n\prime}>0\,,\ U^{n\prime\prime}\leq 0}{\longrightarrow}$ *Social Planner Optimum for some* $\boldsymbol{\eta} > 0$ $\underset{U^{n\prime}>0\,,\ \textit{complete market}}{\longrightarrow}$ *Economic Equilibrium.*

Event tree





At each time $t \in \{0, T\}$, one state $s_t \in 1, \ldots, S$ is realized. Each node $\boldsymbol{\sigma}_t = (s_0, s_1, \ldots, s_t)$ is a "story".

Variables

J securities:

- $x_j(\boldsymbol{\sigma}_t)$ payoff of security j in node $\boldsymbol{\sigma}_t$;
- $p_j(\boldsymbol{\sigma}_t)$ price of security j in node $\boldsymbol{\sigma}_t$.

A portfolio strategy $\mathbf{h} = (\mathbf{h}_0, \ldots, \mathbf{h}_T)$ with $\mathbf{h}_t(\boldsymbol{\sigma}_t) = (h_{t,1}(\boldsymbol{\sigma}_t), \ldots, h_{t,J}(\boldsymbol{\sigma}_t))$ defines the hold in each node, i.e. each possible future (time subscript omitted when understood).

Payoff $\mathbf{z}_{h,p} = \{z(\boldsymbol{\sigma}_t)_{h,p}\}$ of portfolio strategy $\mathbf{h}$ is

$$z(\boldsymbol{\sigma}_t)_{h,p} = \sum_{j=1}^{J} \left(p_j(\boldsymbol{\sigma}_t) + x_j(\boldsymbol{\sigma}_t)\right) h_j(\boldsymbol{\sigma}_t^-) - p_j(\boldsymbol{\sigma}_t) h_j(\boldsymbol{\sigma}_t)$$

where $\boldsymbol{\sigma}_t^-$ is the node preceding $\boldsymbol{\sigma}_t$.

Strategy payoff examples

Buy one unit of asset j at time t^* if the history is $\boldsymbol{\sigma}_t^*$ and sell it at time $t+1$ in any state:

$$\begin{cases} z(\boldsymbol{\sigma}_t) = -p_j(\boldsymbol{\sigma}_t) & \text{if } \boldsymbol{\sigma}_t = \boldsymbol{\sigma}_t^*; \\ z(\boldsymbol{\sigma}_t) = x_j(\boldsymbol{\sigma}_t) + p_j(\boldsymbol{\sigma}_t) & \text{if } t = t^* + 1 \text{ and } \boldsymbol{\sigma}_t^- = \boldsymbol{\sigma}_t^*; \\ z(\boldsymbol{\sigma}_t) = 0 & \text{otherwise.} \end{cases}$$

Buy and hold strategy: buy security 1 ad date zero and keep it until date T

$$\begin{cases} z_0 = -p_1(\boldsymbol{\sigma}_0) \\ z(\boldsymbol{\sigma}_t) = x_j(\boldsymbol{\sigma}_t) \quad \text{for any } t > 0; \end{cases}$$

Asset Span

The asset span $M(p)$ is the set of all possible payoffs:

$$M(p) = \{\mathbf{y} \mid \exists \mathbf{h} \text{ such that } \mathbf{y} = \mathbf{z}_{h,p}\}$$

It generally depends on prices. If S state of nature at each step and T dates then $M(p) \subseteq \mathbb{R}^{S^T}$.

The market is complete if any payoff is possible: $M(p) = \mathbb{R}^{S^T}$.

For instance: $\mathbb{R}^{S^T}$ securities, security $\boldsymbol{\sigma}_t$ pays 1 in $\boldsymbol{\sigma}_t$ and 0 otherwise. The strategy $\mathbf{h}$ is simply to buy an amount $h_{\boldsymbol{\sigma}_t} = y(\boldsymbol{\sigma}_t)$ at date 0.

Payoff Valuation Functional

For the Law of One Price there exists a linear functional $q : M(p) \to \mathbb{R}$, defined for each $\mathbf{y} \in M(p)$ as

$$q(\mathbf{y}) = \{\mathbf{p}_0\mathbf{h}_0 \mid \mathbf{z}_{h,p} = \mathbf{y}\} \ .$$

A strong arbitrage is a portfolio $\mathbf{h}$ such that $q(\mathbf{y}) < 0$ and $\mathbf{z}_{h,p} \geq 0$.

A weak arbitrage is a portfolio $\mathbf{h}$ such that $q(\mathbf{y}) = 0$ and $\mathbf{z}_{h,p} > 0$.

Theorem

There is no strong arbitrage iff q is a positive functional. There is no arbitrage iff q is a strictly positive functional.

Valuation Functional in Complete Markets

Let $e(\boldsymbol{\sigma}_t)$ be the security that pays 1 in node $\boldsymbol{\sigma}_t$ and zero otherwise: Arrow securities.

Any payoff structure $\mathbf{y} = \{y_{\boldsymbol{\sigma}_t}\}$ can be written as a linear combination of Arrow securities $y = \sum_{t,\boldsymbol{\sigma}_t} y_{\boldsymbol{\sigma}_t} e(\boldsymbol{\sigma}_t)$.

The payoff pricing functional is then

$$q(\mathbf{y}) = \sum_{t,\boldsymbol{\sigma}_t} y_{\boldsymbol{\sigma}_t} q(e(\boldsymbol{\sigma}_t)) \,,$$

where $q(e(\boldsymbol{\sigma}_t))$ are the state prices. The gross risk-free interest rate becomes (summing on all nodes at time t)

$$R_f(t) = \sum_{\boldsymbol{\sigma}_t} \frac{1}{q(e(\boldsymbol{\sigma}_t))} \,.$$